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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B.Tech

Semester: VII

Name of the Paper: Non-Conventional Resources and Energy Systems

Paper Code: TEE 702

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	What are solar collectors? Classify the different types of solar collectors and explain their various performance indices.	
(b)	Explain the following. i. Evacuated Tube Collector ii. Flat Plate Collector with Booster Mirrors iii. Compound Parabolic Concentrator iv. Cylindrical Parabolic Concentrator v. Fixed Mirror Solar Concentrator	
(c)	Explain the depletion process of solar radiation as it passes through the atmosphere to reach at the surface of the earth.	
Q2	(20 marks)	CO2, CO6
(a)	Explain the I-V characteristics of a solar cell and define the fill factor. Discuss the significance of the fill factor in the performance of a solar cell.	
(b)	Explain the working principle of solar PV cell. What are the different types of solar PV system	
(c)	Calculate the optimum wavelength of light for photovoltaic generation in a CdS cell. The band gap for CdS is 2.42 eV.	
Q3	(20 marks)	CO4
(a)	Derive the efficiency of a fuel cell.	
(b)	How can hydrogen be stored, and what challenges are faced by fuel cells?	

(c)	Explain the process of producing hydrogen through the following (i) Thermolysis of water (ii) Electrolysis of water	
Q4	(20 marks)	CO3
(a)	Explain the following types of geothermal resources: 1. Hydrothermal Resources 2. Hot Dry Rock Resources 3. Geo-Pressured Resources	
(b)	A 0.6 km thick hot aquifer is located at a depth of 2.5 km and has a porosity of 5 per cent. The density of the under sediments is 3000 kg m^{-3} and its specific heat is $750 \text{ J kg}^{-1} \text{ K}^{-1}$. The temperature gradient in the overlying material is 35°C km^{-1} . Assuming the density and specific heat of water as 1000 kg m^{-3} and $4200 \text{ J kg}^{-1} \text{ K}^{-1}$ respectively, Find the: ● heat content per square kilometer above 45°C and the initial temperature of the aquifer, if the average surface temperature is 12°C	
(c)	What is thermoelectric power converter. Explain the basic principle behind thermoelectric power generator.	
Q5	(20 marks)	
	Explain the following term	CO5
(a)	i. Wind data statistics ii. Capacity factor	
(b)	Following data were measured for a HAWT: Speed of wind = 20 m/s at 1 atm and 27°C Diameter of rotor = 80 m Speed of rotor = 40 rpm Calculate the torque produced at the shaft for maximum output of the turbine.	
(c)	What is a magneto hydrodynamic (MHD) power generator. Explain the basic designs of MHD generator, based on the Faraday principle of operation	